MA385: Tutorial Sheet 1 ANS with some solutions

These exercises are for Tutorial 1 (Week 3). You do not have to submit solutions to these questions. However, similar questions may appear on the assignment, class test, or final exam.

Q1. Let $f(x) = e^x - 2x - 2$.

(a) Show that there is a solution to the problem: *find* $\tau \in [0, 2]$ *such that* $f(\tau) = 0$.

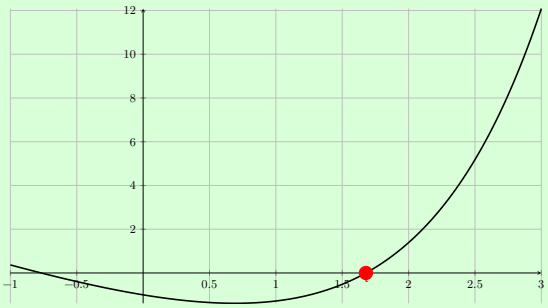
(b) Taking $x_0 = 0$ and $x_1 = 2$, use 6 steps of the bisection method to estimate τ (you may use a computer program to do this, if you like).

(c) Give an upper bound for the error $|\tau - x_6|$ using Theorem 1.2.1.

(d) What is the smallest k needed to ensure that $|x_k - \tau| \leqslant 10^{-4}$?

Answer:

(a) $f(0) = -1 < 0$ and $f(2) \approx 1.389 > 0$. Since f changes sign on $[0, 2]$, and is continuous on $[0, 2]$, by the IVT, there exists τ such that $f(\tau) = 0$.
You don't have to plot the function, but if you did, it would look like this:



(b) I get $\{x_0, x_1, \ldots, x_6\} = \{0, 2, 1, 1.5, 1.75, 1.625, 1.6875\}$.

(c) According to that theorem, $|\tau - x_k| \leqslant (1/2)^{k-1}|b - a|$. Therefore, the error bound is $|\tau - x_6| \leqslant (1/2)^5(2) = 0.0625$. (In practice, the actual error is about 0.031).

(d) Again using that $|\tau - x_k| \leqslant \left(\frac{1}{2}\right)^{k-1}|x_0 - x_1| = \left(\frac{1}{2}\right)^{k-2}$. So we need k large enough so that $(1/2)^{k-2} \leqslant 10^{-4}$. Rearranging, we should see that this means $k \geqslant \log_2(10^4) + 2 \approx 15.288$. So we need to take $k = 16$.

Q2. We wish to estimate $\tau = \sqrt[3]{4}$ numerically by solving $f(x) = 0$ in $[a, b]$ for some suitably chosen f, a and b.

(i) Suggest suitable choices of f, a, and b for this problem.

(ii) Show that f has a zero in $[a, b]$.

(iii) Use 6 steps of the bisection method to estimate $\sqrt[3]{4}$. Again, it is OK to write code to do this for you.

(iv) Use Theorem 1.2.1 to give an upper bound for the error $|\tau - x_6|$.

Answer: Take $f(x) = x^3 - 4$. After that it is pretty similar to the previous question.

Q3. In lectures, the Secant iteration is defined as

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \text{ for } k = 1, 2, 3, \ldots.$$

Show that it can also be written as a *weighted average*

$$x_{k+1} = (1 - \sigma_k)x_k + \sigma_k x_{k-1},$$

and give a formula for σ_k. Derive a formula for σ_k in terms of $f(x_{k-1})$ and $f(x_k)$. Can you give a physical interpretation to σ_k and $1-\sigma_k$ that might justify the claim that the method should be more efficient than bisection?

Answer: You should find that $\sigma_k = \frac{f(x_k)}{f(x_k)-f(x_{k-1})}$.
Justification: Suppose, for example, that $|f(x_{k-1})| \ll |f(x_k)|$ this implies that x_{k-1} is much closer to τ that x_k. Also, this would give that $\sigma_k \frac{f(x_k)}{f(x_k)-f(x_{k-1})} \approx \frac{f(x_k)}{f(x_k)} \approx 1$, and so $x_{k+1} \approx x_{k-1}$. That is, This weights x_{k+1} towards the smaller of $f(x_{k-1})$ and $f(x_k)$.

Q4. Suppose we have a function f with the property that $f(0) = 1$, $f(1) = -1$ and $f(2) = -1$.

(a) f is a quadratic polynomial: find it.

Answer: Write $f(x) = ax^2 + bx + c$. Then solve the three equations $f(0) = 1$, $f(1) = -1$ and $f(2) = -1$ for a, b and c. You should get that $f(x) = x^2 - 3x + 1$.

Show that there is a solution to: *find* $\tau \in [0, 2]$ *such that* $f(\tau) = 0$.

(b) Show that the Secant Method will fail when we take $x_0 = 0$ and $x_1 = 2$.

Answer: The Secant method will always fail if two successive approximations, $xk-1$ and x_k are such that $f(x_{k-1}) = f(x_k)$, because we would be dividing by zero. In the scenario studied here, with $x_0 = 0$ and $x_1 = 2$, we should get $x_2 = 1$. Then we'll have $f(x_1) = f(x_2) = -1$, so the method would fail.

(c) The Secant Method can be *very* sensitive to the choice of x_0 and x_1. By computing a few iterations with $x_0 = 2$ and $x_1 = 0$ show that it appears to converge. (You can write some code to do this, if you like).

Q5. We claimed in class that, if the Secant Method computes the iterates $\{x_0, x_1, x_2 \dots\}$ then

$$|\tau - x_{k+1}| \leqslant C_k |\tau - x_k||\tau - x_{k-1}|.$$

That is, *the error is bounded by the product of the error in the previous two iterates*. As a step towards this justifying this, show that

$$\tau - x_{k+1} = -(\tau - x_k)(\tau - x_{k-1}) \frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]},$$

where

$$f[a, b] := \frac{f(b) - f(a)}{b - a} \qquad \text{and} \qquad f[a, b, c] := \frac{f[b, c] - f[a, b]}{c - a}.$$

Answer: Just algebraic manipulation. For example:

$$\begin{aligned}
(\tau - x_k)(\tau - x_{k-1}) \frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]} &= (\tau - x_k)(\tau - x_{k-1}) \frac{f[x_k, \tau] - f[x_{k-1}, x_k]}{\tau - x_{k-1}} \frac{x_k - x_{k-1}}{f_k - f_{k-1}} \\
&= (\tau - x_k) \left(\frac{f(\tau) - f(x_k)}{\tau - x_k} - \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} \right) \frac{x_k - x_{k-1}}{f_k - f_{k-1}} \\
&= \big(f(\tau) - f(x_k)\big) \big(\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \big) - (\tau - x_k) \\
&= -\tau + x_k - f(x_k) \big(\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \big) = -(\tau - x_{k+1}).
\end{aligned}$$